A Formal Architecture for Regional DLT Settlement Hubs: Resolving the Security Paradox of Convex Monitoring Costs

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Draft: March 31, 2026

Abstract: Central banks face rising cross-border supervision costs as banking groups operate across fragmented regulatory regimes, making traditional oversight mechanisms ineffective and increasing crisis risk. While a distributed ledger technology (DLT) settlement hub offers unified visibility, it introduces a novel Security Paradox: complex regulatory detection methods increase monitoring costs and collateral requirements, thereby risking the exit of compliant banks and systemic instability. This working paper presents a formal architecture for a multi-currency, permissioned regional DLT settlement hub—designed to address the complexities and inefficiencies of cross-border supervision and settlement—that maintains monetary sovereignty across jurisdictions. By adopting a "slashing reserve" model, in which coordinating central banks jointly set monitoring, penalties, transparency, and collateral, regional hub security is decoupled from collateral through efficient detection and a Byzantine-fault-tolerant quorum structure. This design enables capital-efficient, Basel-consistent settlement, ensuring stability and sovereignty, even during crises, and provides a blueprint for modern cross-border financial infrastructure.

JEL Codes: C72, E42, E58, F31, G21

Keywords: Atomic Delivery versus Payment, Distributed Ledger Technology, Central Bank Digital Currency, Cross-Border Supervision, Systemic Risk

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1. Introduction

Central banks face a steadily intensifying challenge: cross-border supervision becomes more complex as banking conglomerates operate across multiple jurisdictions with conflicting laws, incompatible reporting standards, and fragmented regulatory regimes. When a bank in one country transacts with a counterparty abroad, supervisors rarely have full visibility into the underlying risks, collateral positions, or settlement exposures. The core problem is the absence of *consolidated*, timely, machine-verifiable information across jurisdictions. Fragmented ledgers and reporting systems force supervisors to infer group-wide exposures through delayed, partial, and inconsistent snapshots.

When jurisdictions with cross-border conglomerates maintain separate national ledgers, they incur high recurring costs simply to approximate consolidated oversight through bilateral coordination, duplicated audits, and manual reconciliation. Traditional mechanisms, including memoranda of understanding, shared examinations, and periodic reviews, do not scale with the number of entities, products, and cross-border links. They leave large pockets of systemic exposure invisible and unattended until a crisis forces them into view.

The existing system suffers from "fragmented finality," in which cross-border payments depend on slow, opaque correspondent banking chains. This architecture traps liquidity in domestic accounts while counterparties remain exposed during long settlement windows, creating persistent Herstatt risk—the possibility that one party pays while the other defaults before delivering. These vulnerabilities amplify the potential for crisis transmission across national borders. As Adrian et al. (2023) observe, correspondent banking's reliance on bilateral trust inflates transaction fees and widens FX spreads, disproportionately harming developing economies. Under scale, bilateral oversight frameworks collapse, producing high reconciliation costs while leaving systemic risks undetected until they erupt.

To address these structural weaknesses, this research proposes a federated distributed ledger technology (DLT) settlement hub that uses cryptographic lock-and-swap mechanisms to achieve atomic delivery-versus-payment (DvP) across borders.[2] By shifting regional infrastructure from trust-based correspondent banking to programmatically enforced atomic finality, central banks can eliminate principal risk while preserving full national monetary sovereignty. Yet integrating supervisory data across multiple regulatory regimes raises monitoring complexity and inflates collateral requirements beyond Basel III high-quality liquid asset (HQLA) limits, pushing honest validators (banks) out of the system.

The Security Paradox.

A shared ledger improves observability, but it also introduces a new constraint. Integrating supervisory data across heterogeneous regimes can make detection increasingly complex. As monitoring becomes more sophisticated, monitoring costs can become convex and the system

[2] Securities have existed in digital form for some time, so DLT has enabled deterministic settlement finality primarily through programmable liquidity, that is, the tokenization of the 'cash leg' of DvP using wholesale central bank digital currencies (wCBDCs), stablecoins, or tokenized deposits.

substitutes toward collateral. Required reserves can then rise above Basel III high-quality liquid asset (HQLA) limits, pushing honest validators out of the system.

We resolve this paradox by decoupling security from raw collateral through information-efficient detection and Byzantine-fault-tolerant (BFT) guardrails. Permissioned transparency raises baseline detection probabilities and flattens detection elasticity, allowing deterrence to be maintained without capital-inefficient reserve escalation.

1.1 Core Contributions

Permissioned regional DLT hubs directly address the cross-border supervision bottleneck by providing native, machine-readable traceability that eliminates reconciliation costs and delivers baseline detection probabilities unattainable in fragmented legacy systems. Integrating the pain point of costly cross-border supervision (where concerns around trust and efficiency are paramount) with a permissioned transparency layer is the core thesis of the Security Paradox: regional complexity raises monitoring costs, but controlled transparency flattens detection elasticity and restores feasibility.

This paper develops a permissioned DLT settlement hub grounded in distributed systems research (Dwork et al., 1988; Castro and Liskov, 1999). The hub uses permissioned BFT consensus so that any two validator quorums share at least one honest member, supporting deterministic finality under bounded faults. Monetary sovereignty is preserved because each central bank issues and settles only its own wholesale central bank digital currency (wCBDC).

Beyond protocol correctness, the paper integrates game-theoretic incentive design. Using a Stackelberg formulation, a coordinating central bank sets monitoring, penalties, and reserve requirements so that honest validation is strictly optimal for rational participants. The Security Paradox is introduced to show how regional detection elasticity interacts with capital feasibility: when monitoring costs are convex, deterrence can require capital-inefficient collateral unless transparency reduces monitoring complexity.

Finally, the framework aligns DLT security with financial regulation through VaR–HQLA integration. VaR-adjusted slashing reserves connect deterrence to tail risk, while Basel III HQLA ceilings bound collateral feasibility. An “information symmetry coefficient” captures how permissioned transparency raises detection probability at lower institutional cost, allowing information to substitute for idle capital in a regulator-consistent manner.

1.2 The CARICOM Context: Intra-Regional Cross-Border Transaction (In)Efficiency

Why CARICOM.

CARICOM provides a concrete setting in which cross-border financial integration coexists with supervisory fragmentation. The region comprises fifteen small, open economies whose financial sectors are dominated by a small number of cross-border banking groups. In many member states, the top five regional banks account for most system assets, reflecting consolidation and regional expansion by groups headquartered in Trinidad and Tobago, Jamaica, and Canada.

Conglomerate structure and supervisory fragmentation: Lessons from CLICO (2009).

A key lesson for regional oversight is the collapse of Colonial Life Insurance Company (CLICO) in 2009. The distress was driven less by the global crisis shock itself than by structural weaknesses inside the CL Financial group—especially regulatory arbitrage and opaque risk-taking across jurisdictions. CL Financial operated like a leveraged investment vehicle inside an insurance wrapper: it funded long-term, illiquid exposures (including real estate, energy, and higher-risk instruments) with short-term liabilities, creating severe maturity and liquidity mismatches.

Fragmented Caribbean oversight—without effective consolidated supervision of a cross-border conglomerate and with limited group-wide disclosure—allowed leverage and intra-group exposures to accumulate outside any single regulator's full visibility. High-yield products such as "executive flexible premium annuities" reinforced the structure by depending on continuous inflows that were not matched by safely liquid underlying assets. When confidence fell and withdrawals rose, the structure became unstable. Emergency intervention followed: the Government of Trinidad and Tobago announced support and assumed control measures on 30 January 2009, and legislative amendments on 6 February 2009 strengthened intervention powers. Distress then propagated through CLICO's regional subsidiaries across more than 20 Caribbean countries, including Trinidad and Tobago, Barbados, The Bahamas, Guyana, and Jamaica.

These structural weaknesses are compounded by legacy payment and settlement systems, which further constrain intra-regional financial flows. CARICOM economies rely heavily on correspondent banking and extra-regional bridge currencies—primarily the U.S. dollar—for settlement. This dependence produces high transaction fees, wide FX spreads, significant fragmented liquidity trapped in nostro/vostro accounts, and multi-day settlement delays that function as an efficiency tax on trade, remittances, tourism receipts, and interbank flows.

Payments frictions and correspondent banking dependence.

Empirical data show that CARICOM cross-border payments remain among the most expensive globally, with average corridor costs consistently above 6 percent, more than double the G20 and SDG target of reducing remittance costs below 3 percent by 2030 (FSB, 2021). Similar frictions affect trade settlement, where intermediary layering and prefunding requirements inflate costs. When applied to regional trade, services, and remittance flows of US $25–40 billion annually, eliminating these layers could reduce total fees by 40–70 percent and compress FX markups toward interbank levels. Conservative estimates suggest potential annual savings exceeding US $500 million (Statista, 2025; Langrin et al., 2025; OEC, 2024).

These savings form the quantitative base of the cost-benefit model. For a region with a combined GDP of US $130–150 billion, the gains translate into a permanent uplift of roughly 0.6 to 3.0 percent per year, comparable to major trade-liberalization reforms and among the highest-return infrastructure investments available to CARICOM governments. Unlocking liquidity trapped in correspondent banking arrangements would free substantial working capital and improve regional capital efficiency. Although carry costs are difficult to measure precisely, World Bank evidence suggests remittance cost elasticity above one, meaning lower costs stimulate both volume and efficiency.

Implications for hub design.

A regional DLT settlement hub that sits atop multiple, disparate national real-time gross settlement systems (RTGS) within a unified and neutral orchestration layer would fundamentally reshape this landscape by enabling direct peer-to-peer settlement with deterministic finality via atomic swaps. In a region where fragmented national systems routinely delay settlement and generate reconciliation complexity, a shared ledger would achieve machine efficiency while reducing operational risk and capital charges. Immutable transaction records would support real-time automated audits, and programmable compliance, including dynamic credit limits and continuous reporting. This architecture strengthens resilience to operational and regulatory shocks while expanding financial inclusion for smaller member states.

A further advantage of a unified hub is reduced reliance on foreign currency settlement for intra-regional payments. By settling directly on a regional ledger, using a regional token or a basket of wCBDC-linked HQLA, CARICOM members can lower conversion costs and preserve scarce foreign currency liquidity currently immobilized in prefunded correspondent accounts.

The macroeconomic effects extend beyond cost savings. Lower payment frictions would boost employment, SME growth, and productivity—areas the World Bank identifies as structural constraints for small island economies. By reducing fees, FX spreads, and settlement delays, the hub is projected to raise SME profitability and expand regional market participation, generating an estimated 30,000 new jobs annually under conservative assumptions and more than 100,000 with high adoption. Benefits scale nonlinearly: early adoption accelerates network effects, deepening integration, and magnifying returns. When combined with regional CBDCs and tokenized assets, including money market funds and private market vehicles, the hub becomes one of the region's most effective and cost-efficient strategies for employment and growth, with immediate savings that justify investment even under cautious scenarios.[3]

The remainder of the paper is organized as follows. Section 2 specifies the cryptographic architecture and atomic DvP execution. Section 3 introduces the incentive layer as a Stackelberg mechanism with slashing reserves. Section 4 defines detection elasticity and the Security Frontier (and the conditions under which the Insecurity Trap emerges). Section 5 adds VaR-based stochastic robustness and Basel III feasibility. Section 6 provides BFT resilience guarantees for irrational faults. Section 7 concludes with policy implications and an implementation pathway.

2. The Cryptographic Layer: DLT Settlement Hub Architecture

This section describes the "cryptographic layer" of the proposed regional DLT settlement hub: the shared ledger, the assets represented on it, and the mechanisms that deliver verifiable, atomic settlement across jurisdictions. The objective is to specify—at an implementation level—how a hub can (i) shorten cross-border settlement chains, (ii) eliminate principal risk through atomic delivery-versus-payment (DvP), and (iii) produce supervisory visibility as a direct byproduct of a common, uniquely-consistent ledger state.

[3] Regional coordination literature confirms CBDC-induced disintermediation elevates wholesale funding reliance across jurisdictions (Fed, 2024; Aurazo et al., 2024); we extend this to "detection elasticity", which drives collateral beyond L^{max} for regional banking validators.

Permissioned participation and roles.

The hub is permissioned. Validators are known entities—participating central banks and regulated financial institutions (RFIs) such as banks and broker-dealers—admitted under a governance framework. This supports supervisory access, auditability, and high baseline detection probability (a key input to the incentive analysis in Section 3). In contrast to public ledgers, the system's security and compliance posture rests on identified membership, explicit role separation, and policy-configurable monitoring.

Multi-currency and multi-asset representation.

The hub is multi-currency and multi-asset. Central banks issue wholesale central bank digital currency (wCBDC) claims denominated in their sovereign units of account (the "cash leg"). Central securities depositories (CSDs) issue tokenized securities linked to domestic identifiers such as ISINs (the "securities leg"). RFIs hold and transfer these ledger objects subject to protocol rules and jurisdiction-specific constraints. Where useful for transactional efficiency, participants may also utilize a regional bridge asset to reduce repeated FX conversions; however, monetary sovereignty is preserved because each central bank retains issuance and redemption control over its own wCBDC.

System decomposition.

To keep the design modular, the architecture can be read as four components:

1. Participants and roles (who validates, who transacts, who observes).
2. Ledger and state (what is recorded, and what "finality" means in the system).
3. Asset representation (how wCBDC and securities are instantiated as ledger objects).
4. Atomic contract execution (how PvP/DvP is enforced and how failure modes unwind safely).

Ledger integrity and atomicity.

At the base of the hub is an append-only sequence of cryptographically linked blocks. Each block header includes the block index, prior hash, timestamp, and a Merkle root summarizing the transactions included in the block. This construction makes the ledger tamper-evident: any modification to historical records changes the corresponding hash and breaks hash-chaining, which can be immediately detected by validators that maintain the canonical chain state.

The hub's cryptographic primitives serve three system-level functions, used throughout the remainder of the paper:

- **Integrity and attribution:** transactions are digitally signed, binding actions to known identities.
- **Consistency:** all honest participants converge on a single ledger state, allowing a unified view of balances, escrows, and contract states across jurisdictions.
- **Atomicity:** settlement can be expressed as "commit-or-abort" state transitions, eliminating partial settlement and principal risk.

2.1 Atomic DvP Smart Contract Engine

A shared settlement hub provides multi-jurisdictional supervisory visibility as a structural consequence of atomic execution: cross-border payments, collateral postings, escrow states, and validator votes are recorded on a single ledger under a single state-transition rule. This collapses reconciliation latency and lowers monitoring overhead, reducing the need to reconstruct exposures from fragmented national ledgers.

Transaction safety requirements.

Each transaction that updates the hub's global ledger must satisfy three safety requirements:[4]

1. **Authenticity**: the transaction is digitally signed and verifiable against the initiator's public key.
2. **Conservation**: inputs equal outputs for each currency and security type (no creation or destruction of units).
3. **Authorization**: inputs are controlled by the signers and satisfy all spending conditions.

DvP execution sequence.

At the hub's core is an atomic DvP smart contract engine that enforces a strict commit-or-abort settlement rule: either both legs execute together, or neither executes. A transaction therefore, cannot partially settle. The protocol proceeds in four stages:

1. Instruction submission. Buyer and seller submit matching hashed instructions (quantities, instruments, FX rates, and compliance flags).
2. Two-leg escrow. If instructions match, assets are escrowed on both the cash ledger and the securities ledger.
3. Validation and consensus. Validators run a consensus round to confirm escrows are funded, signatures are valid, and protocol rules are satisfied.
4. Finalization or unwind. Upon supermajority quorum, settlement finalizes (wCBDC transfers to the seller and the tokenized security to the buyer). If consensus fails or a timeout occurs, the escrow is cancelled and assets revert to their original owners, preventing liquidity traps.

The global ledger applies updates using a deterministic transition function, enforcing strict atomicity.

3. The Incentive Layer: A Stackelberg Formulation

This section specifies the incentive layer of the DLT settlement hub: the economic mechanism that makes honest validation a best response for rational participants, even when cross-border regulatory arbitrage opportunities exist. The mechanism is based on a slashing reserve design (Buterin and Griffith, 2017). Validators post a reserve that can be forfeited ("slashed") upon detected deviation, and where the coordinating central bank(s) choose monitoring and enforcement parameters to sustain honesty in equilibrium.

The modeling objective is operational: we make explicit (i) who moves when, (ii) what information is available at each decision point, and (iii) which parameters central banks can adjust as volumes and supervisory complexity evolve. We therefore model the hub as a repeated Stackelberg game in which coordinating central bank(s) act as leaders setting policy, and validators act as followers choosing whether to validate honestly or to deviate.

3.1 Economic mechanism: slashing reserve and equilibrium logic

The hub's stability rests on aligning validator incentives with ledger integrity through a self-enforcing economic mechanism that remains effective under regional liquidity fragmentation and regulatory divergence. Rather than assuming trust among institutions across jurisdictions, the mechanism treats validation as an economically motivated action: validators participate if the present value of honest validation dominates the one-shot gains from deviation.

Core idea. The hub implements a *slashing reserve* model: validators earn a recurring stream of validation-related revenues when they behave honestly, but face an expected loss (reserve forfeiture and exclusion from future rewards) if they deviate and are detected. Deterrence is therefore a function of:

- the **one-shot gain** from deviation,
- the **probability of detection**, and
- the **severity of punishment** (including slashing and loss of continuation value).

This section makes these objects explicit and links them to the central bank's policy instruments.

For clarity, the incentive layer can be decomposed into five model ingredients:

1. **Mechanism**: rewards for honest validation and slashing/exclusion upon detected deviation.
2. **Players**: leader (coordinating central bank(s)) and followers (validators).
3. **Timing**: policy announcement, validator action, detection/punishment realization, continuation.
4. **Payoffs**: a discounted stream of legitimate rewards under honesty versus a one-shot illicit gain under deviation.
5. **Equilibrium concept**: a Stackelberg (subgame-perfect) equilibrium implementing honesty as the follower best response.

Section structure. Section 3.2 defines the leader's objective and the within-period timing. Section 3.3 derives validator utilities and the incentive-compatibility condition (the "Honesty Invariant"). Section 3.4 explains why regulatory arbitrage creates a weakest-link constraint that can bind system-wide feasibility.

3.2 Central Bank Objective and Stackelberg Formulation

Central banks must adjust policy dynamically across jurisdictions with heterogeneous market conditions and transaction volumes spanning settlement and regional cross-border payments. To capture this regional scale and complexity, we cast the DLT hub as a dynamic mechanism design

problem with Stackelberg structure: coordinating central bank(s) choose policy instruments first, and validators from participating regulators and RFIs respond optimally, including in the presence of cross-jurisdiction arbitrage opportunities.

We model a two-level game repeated over an infinite horizon. Within each period: (1) central banks announce policy vector θ across jurisdictions spanning complex cross-border banking conglomerates, and (2) validators observe θ and simultaneously choose $a \in \{H, D\}$. Payoff primitives are: honest validators (H) collect a perpetual stream of local and external transaction revenues represented by a per-period reward $R > 0$, discounted by a factor $\beta \in (0,1)$. Deviators (D) obtain a one-shot illicit gain G, but face detection with probability $\rho(\theta) \in [0, 1]$. Upon detection, a deviator incurs a penalty $P = \pi L$, $\pi \in (0,1]$. This refers to the reduction ("slashing") of posted collateral L and permanent exclusion from future validator rewards with zero continuation value, where the maximum potential penalty $\pi = 1$ reflects the "grim-trigger" punishment (Fudenberg and Tirole, 1991).[4]

3.3 Validator Utility and Incentive Compatibility

This subsection states the validator's payoff from honest validation versus deviation and derives the incentive compatibility (IC) condition that sustains honesty as a best response.

An honest validator earns an infinite-horizon stream of rewards. The next display gives the corresponding expected utility. The infinite-horizon honest validator utility across jurisdictions is:

$$U^H = \frac{R(\theta)}{1 - \beta}$$

and the deviation utility becomes:

$$U^D(\theta) = \big(1 - \rho(\theta)\big)G(\theta) - \rho(\theta)\pi L.$$

Honesty Invariant (incentive compatibility)

The hub is incentive compatible if, for every validator and for the policy vector chosen by the leader, honest validation weakly dominates deviation, ruling out profitable cross-border regulatory arbitrage. We capture this requirement with the "Honesty Invariant":

$$U^H \geq U^D.$$

For any announced $\theta = (\rho, \pi, L)$, the validator's best response H is when IC holds strictly, and D when IC is violated and deviation is profitable. Validators best-respond each period, while the leader's forward-looking choice of θ keeps honesty as a subgame-perfect equilibrium over an infinite horizon of regional interaction.

Leader's problem

[4] Even if central banks are the sole validators, RFIs still rationally exit once the IC constraint implies negative economic utility.

Given that deviations are ruled out on the equilibrium path when IC holds, the coordinating central bank chooses policy to minimize its objective subject to the IC constraint. The resulting constrained problem is:

$$\min_{\theta} C(\theta) + \pi L - E(\theta)$$

subject to $U^H(\theta) \geq U^D(\theta)$.

The Stackelberg formulation transforms this constraint into a dynamic policy rule: as transaction volumes increase or or the gains from deviation increase, the coordinating central bank updates θ_t to remain on or above the Security Frontier, and an equilibrium is a pair (θ_t^*, a_t^*) such that, validators play $a_t^* = H$ because IC, $U^H(\theta_t^*) \geq U^D(\theta_t^*)$, holds (follower optimality) and θ_t^* solves the central bank's constrained problem among all policies that implement H as a best response (leader optimality).

Coordinating central banks internalize three components: (1) security cost of monitoring and enforcement, $C(\theta_t)$, (2) systemic loss (πL_t) from successful cross-border deviation (reputational damage, contagion, liquidity stress), and efficiency gains $E(\theta_t)$ from stable regional operations (lower trapped liquidity, improved FX conditions).

Under standard convexity conditions, the Lagrangian characterizes the "optimal enforcement frontier" that links L_t to dynamic deterrence needed via ρ and π. By backward induction, the resulting "systemic honesty" for all rational jurisdictions is the unique sub-game perfect equilibrium (SPE) penalty threshold πL_t^* that ensures region-wide honesty when central banks enforce the honesty invariant across regional operating conditions through issuance control (affecting R and G) and collateral scaling. Thus, the central bank endogenously sets the Stackelberg pair $\theta_t^* = (\rho_t^*, L_t^*)$ to satisfy the strict inequality $U^H(\theta) > U^D(\theta)$ in period 1.

3.4 Regulatory Arbitrage

Regulatory arbitrage arises when an actor exploits discrepancies across national oversight frameworks to increase the one-shot gain from deviation while lowering the effective probability of detection. In a federated hub, an arbitrageur targets the jurisdiction where monitoring is least effective on the margin—i.e., where increasing detection is most costly and where enforcement is weakest.

In this model, arbitrage is not merely a legal loophole. It is a mechanism-design constraint: because the hub must satisfy IC for all participating validators, the jurisdiction with the weakest enforcement environment effectively determines the minimum deterrence required system-wide.

Operationally, this produces a "weakest-link" logic:

- If one jurisdiction has low detection probability or highly elastic (costly) monitoring, the leader must compensate via higher punishment or higher required reserve.
- That higher reserve applies broadly and can become capital-inefficient for otherwise compliant validators in stronger jurisdictions.
- If reserve demands breach prudential constraints, compliant participants rationally exit, reducing validator diversity and increasing systemic fragility.

Regulatory arbitrage opportunities exist and stem from exploiting structural discrepancies between national oversight frameworks to maximize G while minimizing $\rho(\cdot)$. This paper treats regulatory arbitrage not merely as a legal loophole, but as a dynamic systemic risk that dictates the hub's capital requirements. In a federated hub, arbitrageurs target jurisdictions with the lowest detection probability and the highest detection elasticity, where the marginal cost of tightening monitoring is greatest. This creates a "tail risk" scenario, where a single successful breach in a low-supervision jurisdiction can result in a systemic loss that far exceeds the average transaction value. Because the hub must satisfy the IC constraint for all members, the "weakest" jurisdiction's detection elasticity effectively sets the floor for the regulatory reserve requirement for the entire network, which we call the capital-feasible operating region. If one jurisdiction's detection elasticity is too high, it forces the hub to demand a level of reserve that is capital-inefficient for honest banks in "stronger" jurisdictions.

4.0 Detection Elasticity and the Security Paradox

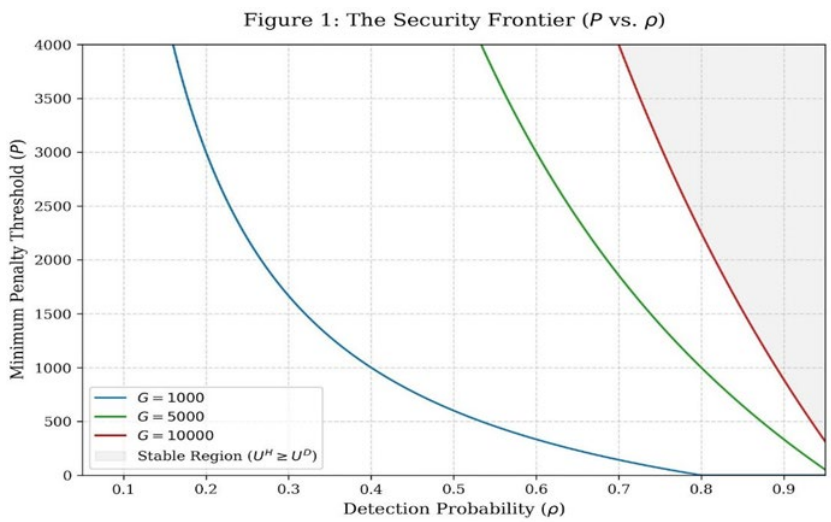
This section formalizes detection elasticity and uses it to define the paper's central mechanism-design problem: the Security Paradox. The paradox is that cross-jurisdiction supervision can become so complex that improving detection requires disproportionately costly monitoring. The leader then substitutes away from monitoring and toward collateral, potentially pushing required reserves above prudential bounds and triggering exit by honest participants.

Regulatory arbitrage validates the Security Paradox. Arbitrage creates an "arbitrage premium" (extra illicit gain in a weakly supervised jurisdiction). Deterring that premium may require a slashing reserve so large that it is not capital-feasible. When the required reserve breaches prudential limits, the hub enters an Insecurity Trap: deterrence can only be restored by collateral levels that force compliant validators to exit.

Security Frontier

Figure 1 illustrates the Security Frontier: the boundary in θ_t-space where validators spanning jurisdictions are indifferent between honesty and deviation. Inside the feasible region, validators strictly prefer H over D despite cross-border arbitrage temptations. As cross-border volumes and complexity increase, the potential gain rises. To preserve incentive compatibility, central banks must respond by increasing detection probability (monitoring) and/or increasing the expected penalty (slashing), which often translates into a higher required reserve. The slashing reserve is "skin in the game": the minimum capital a validator must lock in escrow with the central bank, calibrated to regional HQLA constraints.

The key design implication is that the hub must implement permissioned transparency (baseline monitoring) across all jurisdictions to flatten the detection-elasticity curve and reduce the incentive for arbitrage in the weakest-link jurisdiction.



4.1 Monitoring Cost Convexity: The Cost Function of Detection

The Stackelberg equilibrium in Section 3 allowed for a generic monitoring-cost object. We now make monitoring technology explicit. The purpose is to show why cross-border supervisory complexity can generate *strictly convex* monitoring costs and why this convexity is the source of the Security Paradox.

In a complex, multi-jurisdictional environment, the central bank's monitoring cost is strictly convex due to the complex "compliance mesh". Regional monitoring adopts an isoelastic form:

$$C(\theta_t) = \alpha \frac{\rho^{1+\varepsilon}}{1+\varepsilon}, \;\; \alpha > 0,$$

where ε is the detection elasticity parameter that measures how sharply marginal monitoring costs rise as detection probability increases. Differentiating, $C'(\theta_t) = \alpha\rho^{\varepsilon}$, the marginal monitoring cost increases polynomially in ρ, and convexity holds since $C''(\theta_t) = \alpha\varepsilon\rho^{\varepsilon-1} > 0$. The limit as detection elasticity vanishes is: $lim_{\varepsilon\to 0} C(\theta_t) = \alpha\rho$. So, $\varepsilon \to 0$ corresponds to linear, scalable monitoring. As ε rises, marginal cost scales faster, consistent with costly, multi-jurisdictional monitoring, where each incremental unit of oversight becomes exponentially expensive.

4.2 Information Symmetry and Mapping

This subsection introduces a policy-relevant lever: permissioned transparency. We capture its effectiveness using the "information symmetry coefficient", $\sigma \in (0,1]$. Intuitively, measures how much cross-jurisdiction supervisory information is effectively shared and usable for detection, relative to a fully integrated benchmark. Higher shifts the monitoring technology outward: the leader can achieve a given detection probability at lower institutional cost.

Two limiting cases are useful for interpretation:

- ("Perfect symmetry"): supervisors have near-complete, interoperable visibility, so detection can approach certainty for relevant behaviors (i.e., $\rho(1,\varepsilon) = 1^{1/1+\varepsilon} = 1$).

- ("High opacity"): supervisors have little effective visibility into local ledgers, so detection probability falls toward zero absent additional enforcement capacity.

Let the detection probability entering the central bank's incentive constraint be $\rho(\sigma, \varepsilon)$, with standard properties $\rho'(\cdot) > 0$, $\rho''(\cdot) < 0$. Assuming $\pi = 1$, the IC constraint for validator i is:

$$U^H \geq \big(1 - \rho(\sigma, \varepsilon)\big)G - \rho(\sigma, \varepsilon)L,$$

The introduction of σ allows the central bank (the leader) to achieve ρ^* at a lower institutional cost. This shift effectively "subsidizes" the security of the network, leading to a new equilibrium reserve. Solving for the implied slashing reserve yields:

$$L^*(\sigma) = \frac{U^D - U^H}{\rho^*(\sigma, \varepsilon)} = \frac{\big(1 - \rho^*(\sigma, \varepsilon)\big)G}{\rho^*(\sigma, \varepsilon)}$$

Hence, increasing $\rho(\sigma, \varepsilon)$ (better detection) lowers L^*, but doing so entails cost $C\big(\rho(\sigma, \varepsilon)\big)$ that grows convexly in the endogenous detection probability ($\rho(\sigma, \varepsilon)$). As σ increases, ρ rises because it is cheaper to implement.

To illustrate how information symmetry impacts regulatory effectiveness and systemic resilience, consider the following. When $\sigma = 1$, the leader has high-quality visibility, and detection becomes reliable. When $\sigma \approx 0$, information is fragmented and detection falls toward zero.

The relationship is mediated by ε, the complexity parameter. As jurisdictions fragment and complexity increases, the "returns" to transparency diminish: more permissioned transparency is required to achieve the same effective detection.

In equilibrium, higher effective detection reduces the required reserve: L^* falls, freeing bank capital for lending and market-making. The Security Paradox is the converse: if permissioned transparency is too low (low σ), then L^* must rise to maintain deterrence. If that implied reserve crosses the HQLA ceiling, the system becomes legally or economically infeasible.

The determination of P is crucial for the regional hub. If P is harmonized such that regulatory arbitrage in one country results in an automatic loss of operating license in all others, the value of P skyrockets. This allows the hub to remain secure even if σ is temporarily low, because the fixed cost of betrayal is too high for any rational bank to risk.

4.3 The Substitution Effect: Optimal Monitoring under Convex Costs

At this stage, the leader's Stackelberg problem can be read as a trade-off between (i) raising detection through monitoring, which is institutionally costly under convexity, and (ii) raising the expected penalty through higher required collateral, which is capital-inefficient for honest validators.

The central bank's Stackelberg optimization trades off monitoring cost and required collateral:

$$\min_{\rho \in (0,1)} C(\rho) + \lambda L^*(\rho) = \min_{\rho} \left[\alpha \frac{\rho^{1+\varepsilon}}{\sigma(1+\varepsilon)} + \lambda \frac{G - \frac{R}{1-\beta}}{\rho\pi} \right]$$

where $\lambda L > 0$ is the shadow price on validator capital efficiency.

Define $\Delta \equiv G - R/1 - \beta > 0$ as the *net temptation*: the one-shot illicit gain net of the honest continuation value. This term summarizes how attractive deviation is on the margin. Expressed in terms of net temptation, the leader's problem becomes:

$$\min_{\rho} \alpha \frac{\rho^{1+\varepsilon}}{\sigma(1+\varepsilon)} + \lambda \frac{\Delta}{\pi} \frac{1}{\rho}$$

The first-order condition yields the "Optimal Monitoring Intensity":

$$\rho^*(\varepsilon, \sigma) = \left(\frac{\lambda \Delta \sigma}{\alpha \pi}\right)^{\frac{1}{2+\varepsilon}}$$

and the implied collateral requirement:

$$L^*(\varepsilon, \sigma) = \frac{\Delta \sigma}{\rho^*(\varepsilon, \sigma)\pi} = \frac{\Delta \sigma}{\pi} \left(\frac{\alpha \pi}{\lambda \Delta}\right)^{\frac{1}{2+\varepsilon}}$$

4.4 Comparative statics: Deriving the Optimal Monitoring ρ^*

We now summarize how the Stackelberg-optimal detection choice responds to changes in (i) regional monitoring complexity and (ii) permissioned transparency. We focus on the empirically relevant feasible region where the shadow price of capital and the net temptation are high relative to baseline monitoring costs. In this region, the optimal detection probability takes the reduced-form expression below.

Comparative-statics summary.

- Higher regional complexity makes marginal monitoring more expensive, pushing the leader toward lower optimal detection and higher required collateral.
- Higher information symmetry makes detection cheaper/more effective, allowing higher optimal detection and lower required collateral.

$$\rho^*(\varepsilon, \sigma) = K^{\frac{1}{2+\varepsilon}}; \quad L^*(\varepsilon, \sigma) = \frac{\Delta}{\pi} K^{-\frac{1}{2+\varepsilon}}.$$

where $K = \lambda \Delta \sigma / \alpha \pi > 1$.

Lemma 1 (Monotonicity of Slashing Reserve). *Assume $K > 1$. Then the optimal detection probability, ρ^* is strictly decreasing in regional complexity ε, while the equilibrium slashing reserve L^* is strictly increasing in ε.*[5] *Simultaneously, ρ^* is strictly increasing and L^* is strictly decreasing in information symmetry σ.*[6]

The proof proceeds by differentiating the closed-form expressions for optimal detection and the implied reserve, using the convexity restrictions introduced earlier.

Since $K > 1$, the factor $K^{1/(2+\varepsilon)}$ is strictly decreasing in ε, so ρ^* decreases with regional complexity. Because

[5] Complexity acts as a "liquidity tax," forcing the leader to rely on larger ex-ante collateral.
[6] Information symmetry recovered through the regional "compliance mesh" directly substitutes for idle capital.

$$L^*(\varepsilon,\sigma) = \frac{\Delta}{\pi} K^{-\frac{1}{2+\varepsilon}},$$

the reserve inherits the opposite monotonicity and is strictly increasing in ε. Differentiating yields

$$\frac{\partial L^*}{\partial \varepsilon} = \frac{\Delta}{\pi} K^{-\frac{1}{2+\varepsilon}} \cdot \frac{\ln K}{(2+\varepsilon)^2} > 0,$$

where the inequality follows from $K > 1$. Since K is increasing in σ, ρ^* is strictly increasing in information symmetry. And because L^* is decreasing in K, it is strictly decreasing in σ. Therefore, the lemma follows. □

Following Bolton and Dewatripont (2005), Lemma 1 formalizes the substitution effect: when monitoring technology becomes more convex, the leader substitutes away from costly ex-post detection and toward ex-ante collateral. In this model, the channel runs through θ_t, linking detection elasticity directly to capital requirements in a central-bank-led hub.

Definition 1 (Security Paradox/Insecurity Trap). Let L^{max} denote the maximum required collateral per validator under Basel III liquidity rules that is admissible in the feasible region. The Insecurity Trap is defined as the set: $\mathcal{T} = \{(\sigma,\varepsilon): L^*(\sigma,\varepsilon) > L^{max}\}$. In this region, compliant RFIs cannot participate honestly as required reserves exceed regulatory prudential limits.

Proposition 1 (Existence of the Security Paradox). *There exist economically relevant parameter values generating the Insecurity Trap in each of the following cases: (a) the complexity trap, where sufficiently high regional complexity ε forces $L^*(\varepsilon,\sigma) > L^{max}$; and (b) the opacity trap, where sufficiently low information symmetry σ forces $L^*(\varepsilon,\sigma) > L^{max}$.*

By Lemma 1, the equilibrium reserve $L^*(\varepsilon,\sigma)$ is strictly increasing in regional complexity ε and strictly decreasing in information symmetry σ. Consider first case (a). Fix any admissible σ. Since $L^*(\varepsilon,\sigma)$ is continuous and strictly increasing in ε, and since there exists a low-complexity benchmark at which $L^*(\varepsilon,\sigma) \leq L^{max}$ while at sufficiently large ε the reserve exceeds L^{max}, the Intermediate Value Theorem implies that there exists ε_0 such that

$$L^*(\varepsilon_0,\sigma) = L^{max}.$$

Hence for all $\varepsilon > \varepsilon_0$, strict monotonicity implies $L^*(\varepsilon,\sigma) > L^{max}$. This establishes the complexity trap.

For case (b), fix any admissible ε. Since $L^*(\varepsilon,\sigma)$ is continuous and strictly decreasing in σ, and since totally opaque systems imply arbitrarily high monitoring cost and therefore arbitrarily large required reserve, there exists a low-information-symmetry benchmark σ_0 such that

$$L^*(\varepsilon,\sigma_0) = L^{max}.$$

By strict monotonicity, for all $\sigma < \sigma_0$,

$$L^*(\varepsilon,\sigma) > L^{max}.$$

Thus, the opacity trap also generates the Insecurity Trap. Therefore, the set $\mathcal{T}$ is nonempty whenever the parameter region includes sufficiently high complexity or sufficiently low information symmetry. □

In this regime, honest validators cannot post the required collateral: (a) beyond ε_0, the set of honest participants must exit the system, and (b) below σ_0, honest participation is legally or economically impossible. This proof formalizes the substitution effect: increased transparency and reduced complexity through a central-bank-led DLT hub shift enforcement away from expensive ex-ante collateral toward efficient ex-post detection. This mechanism resolves the Security Paradox, unlocking a welfare-maximizing "liquidity dividend" while ensuring systemic honesty remains the unique subgame-perfect equilibrium for all participating jurisdictions.

5. Stochastic Robustness: VaR-HQLA Integration

This section extends the deterministic incentive analysis in Sections 3–4 to a **stochastic** environment. The objective is to integrate (i) tail risk in illicit gains from cross-border regulatory arbitrage with (ii) prudential liquidity constraints faced by validator banks. The result is a **non-empty regulatory feasibility region**: a set of parameters in which honesty can be sustained at a chosen confidence level without forcing compliant participants into the Insecurity Trap.

In practice, the one-shot gain from deviation is not constant. It varies with settlement volumes, market stress, and temporary enforcement gaps. A hub that is incentive compatible "on average" may still be vulnerable in the tail. We therefore impose a probabilistic honesty requirement that matches standard value-at-risk (VaR) conventions used in banking supervision.

We treat illicit gain as a random variable and choose a tolerance level for "profitable deviation" events. This yields a VaR-adjusted collateral floor that can be compared directly to Basel III high-quality liquid asset (HQLA) limits.

Section 5.1 derives a probabilistic honesty invariant and a VaR-adjusted reserve floor (Lemma 2). Section 5.2 combines that floor with the Stackelberg-optimal reserve and the HQLA ceiling to define the regulatory feasibility region that must be nonempty for the hub to be deployable at scale.

5.1 Probabilistic Honesty Invariant

We model the illicit gain from deviation as stochastic $G \sim N(\mu_G, \sigma_G^2)$ to reflect volatility in high-value cross-border settlement opportunities. Let the regulator tolerate at most α probability that deviation is profitable. We align this supervisory tolerance with the VaR convention by setting $\alpha = \rho$, the honesty confidence level.

With this convention, the probabilistic honesty requirement can be stated as a tail event restriction: profitable deviation must occur only in the far tail of the illicit-gain distribution. On deviation, the validator earns $U^D(G) = (1-\rho)G - \rho\pi L$. A probabilistic honesty invariant therefore requires the tail probability bound $\alpha \in (0,1)$, which is equivalently written as $\Pr[\, G \leq (U^H + \rho\pi L)/1-\rho \,] \geq 1-\alpha$:

Lemma 2: VaR-Adjusted Collateral Floor. *Suppose illicit gains are stochastic and the regulator tolerates at most probability* $\alpha \in (0,1)$ *that deviation is profitable. Under the tail-risk approximation* $(1-\rho) \approx 1$ *and* $U^H \approx 0$ *in the far tail, the probabilistic incentive-compatibility constraint is satisfied if and only if the slashing reserve* L *satisfies*

$$L^* \geq L^{VaR}(\rho) = \frac{\mu_G + z_{1-\alpha}\sigma_G}{\rho\pi},$$

where $z_{1-\alpha}$ denotes the $(1-\alpha)$-quantile of the standard normal distribution and $\rho\pi$ is the effective penalty rate.

The proof rewrites "profitable deviation" as a tail event and then applies a standard normal-quantile representation to obtain a closed-form reserve floor.

Let deviation be profitable when the deviation payoff weakly exceeds the honest continuation payoff. By the model definition, the deviation payoff is

$$U^D(G) = (1-\rho)G - \rho\pi L.$$

The probabilistic honesty condition requires

$$\Pr[U^D(G) \leq U^H] \geq 1-\alpha.$$

Substituting $U^D(G)$ and rearranging gives

$$\Pr[(1-\rho)G \leq U^H + \rho\pi L] \geq 1-\alpha.$$

Under the tail-risk approximation $(1-\rho) \approx 1$ and $U^H \approx 0$, this becomes

$$\Pr[\, G \leq \rho\pi L \,] \geq 1-\alpha.$$

Since $G \sim N(\mu_G, \sigma_G^2)$, the standardized variable

$$Z = \frac{G-\mu_G}{\sigma_G}$$

is standard normal. Therefore,

$$\Pr\left[\frac{G-\mu_G}{\sigma_G} \leq \frac{\rho\pi L - \mu_G}{\sigma_G}\right] \geq 1-\alpha.$$

By the definition of the $(1-\alpha)$-quantile, this is equivalent to

$$\frac{\rho\pi L - \mu_G}{\sigma_G} \geq z_{1-\alpha}.$$

Rearranging yields

$$L \geq \frac{\mu_G + z_{1-\alpha}\sigma_G}{\rho\pi}.$$

Hence the probabilistic incentive-compatibility constraint holds if and only if the slashing reserve lies above the VaR-adjusted collateral floor. [7] □

5.2 Regulatory Feasibility Region and Basel III Alignment

We now combine tail-risk deterrence (Section 5.1) with the deterministic Stackelberg solution (Section 4) and with prudential liquidity limits. This yields the Regulatory Feasibility Region: the set of parameter values under which the hub can implement honesty at the chosen VaR confidence level without requiring collateral in excess of Basel III HQLA capacity.

Feasibility constraints.

[7] This floor ensures that expected penalties cover extreme tail risks in illicit gains, effectively mirroring the market risk capital requirements found in Basel III standards.

In this model, feasibility is governed by three constraints that must hold simultaneously for the equilibrium slashing reserve:

- **Stochastic deterrence floor (VaR).** The reserve must exceed the VaR-adjusted collateral floor from Lemma 2.
- **Deterministic incentive compatibility (Stackelberg).** The reserve must be at least as large as the Stackelberg-implied requirement from Lemma 1 under monitoring convexity.
- **Prudential liquidity ceiling (Basel III).** The reserve must not exceed the HQLA ceiling admissible for validators.

Together, these bounds define a three-way feasible set:

$$L^{VaR}(\theta_t) \leq L^*(\theta_t) \leq L^{HQLA}$$

Policy calibration over the cycle.

The permissioned design of the regional hub allows the policy authority to recalibrate deterrence as market conditions change.

Stress periods. When tail risk rises (larger extreme realizations of illicit gain), the central bank can increase the effective penalty rate π_t (macro-prudential policy parameter), which tightens the probabilistic honesty invariant $L^*(\theta_t) \geq L^{VaR}(\theta_t)$. Intuitively, harsher slashing compensates for higher tail gains without requiring a proportionate increase in posted reserves.

Normal periods. When tail risk recedes, required collateral can relax toward the deterministic IC optimum $L^*(\theta_t)$, preserving capital efficiency while remaining above the VaR-adjusted floor.

Section 6 complements this economic feasibility region with structural Byzantine-fault-tolerant guardrails that protect the ledger even when some nodes are compromised or behave irrationally.

The Stackelberg mechanism delivers endogenous risk controls through economic parameters rather than exogenous administrative caps. In contrast to traditional centralized settlement (Federal Reserve, 2024), which relies on three exogenous stability pillars—(1) quantity caps preventing deposit displacement, (2) tiered remuneration discouraging hoarding, and (3) Liquidity Coverage Ratio (LCR) calibration preserving HQLA—the DLT protocol embeds these functions directly into the incentive structure.

Proposition 2: Regulatory Arbitrage Elimination. *Let $L_j^*(\varepsilon_j, \sigma)$ denote the equilibrium slashing reserve for a validator in jurisdiction j. Let L^{min} denote the globally sufficient threshold required to maintain the regional honesty invariant across all jurisdictions. Regulatory arbitrage is eliminated if*

$$L_j^*(\varepsilon_j, \sigma) \geq L^{min} \text{ for all } j.$$

This ensures that while local reserve requirements may vary with jurisdictional complexity ε_j, all validators meet the minimum threshold necessary to align distributed settlement costs with centralized prudential standards.

Regulatory arbitrage exists if there are jurisdictions j and k such that $L_j^*(\varepsilon_j, \sigma) < L_k^*(\varepsilon_k, \sigma)$, creating an incentive for institutions to relocate to the jurisdiction with lower reserve

requirements. By enforcing a hub-wide protocol that rejects any validator with $L_j^*(\varepsilon_j, \sigma) < L^{\min}$, the regulator standardizes the minimum expected penalty across all jurisdictions. By Lemma 1, $L_j^*(\varepsilon_j, \sigma)$ is strictly increasing in local complexity ε_j and strictly decreasing in information symmetry σ. The Stackelberg mechanism endogenously delivers $L_j^* \geq L^{\min}$ through economic parameters rather than exogenous administrative caps. Since the protocol rejects any $L_j^* < L^{\min}$, the "race to the bottom" is structurally impossible, and economic equivalence is maintained because the deterrent effect is identical across borders. □

Proposition 3 (Basel-Consistent Robust Security). *Fix a confidence level $\rho \in (0,1)$ consistent with Basel III tail-risk standards. Suppose a policy satisfies:*

1. *the stochastic incentive-compatibility floor $L^* \geq L^{VaR}(\rho)$, where $L^{VaR}(\rho)$ is the VaR-adjusted collateral threshold from Lemma 2; and*
2. *the prudential liquidity ceiling $L^* \leq L^{max}$, where L^{max} is the maximum collateral admissible under the relevant Basel III HQLA constraint.*

Then the policy implements a Basel III-consistent robust security regime: validators remain honest with probability at least ρ, and collateral remains within prudential limits.

Condition (1) implies that the probabilistic incentive-compatibility constraint holds at confidence level ρ. By Lemma 2, any reserve satisfying $L^* \geq L^{\mathrm{VaR}}(\rho)$ ensures that deviation is unprofitable except with tail probability at most $1 - \rho$. Hence honest behavior is sustained with probability at least ρ. Condition (2) guarantees that the required collateral does not violate the prudential ceiling imposed by the Basel III liquidity framework. Therefore, the reserve is both sufficiently large to deter deviation and sufficiently small to remain feasible under prudential regulation. Together, these two conditions imply that the system achieves robust security without triggering the Security Paradox of Proposition 1. Hence the policy is Basel III-consistent. □

Proposition 4 (Nonemptiness of the Regulatory Feasibility Region). *The regulatory feasibility region*

$$\mathcal{F} = \{(\varepsilon, \sigma): L^{VaR}(\rho) \leq L^*(\varepsilon, \sigma) \leq L^{max}\}$$

is nonempty if and only if there exists at least one admissible parameter pair (ε, σ) such that

$$L^{VaR}(\rho) \leq L^{max}.$$

By definition, $\mathcal{F}$ is nonempty if and only if there exists at least one admissible pair (ε, σ) for which the equilibrium reserve satisfies both

$$L^*(\varepsilon, \sigma) \geq L^{\mathrm{VaR}}(\rho) \text{ and } L^*(\varepsilon, \sigma) \leq L^{max}.$$

This is possible only if the lower bound does not exceed the upper bound, i.e.

$$L^{\mathrm{VaR}}(\rho) \leq L^{max}.$$

If $L^{\mathrm{VaR}}(\rho) > L^{max}$, then no reserve can simultaneously satisfy the honesty floor and the prudential ceiling, so the feasibility region is empty. Since $L^{\mathrm{VaR}}(\rho)$ is decreasing in information symmetry σ through the equilibrium mapping in earlier propositions, a sufficiently high σ can depress the reserve enough to make the region nonempty. Thus, the region exists only when

informational conditions are strong enough to reconcile prudential feasibility with incentive compatibility. □

Proposition 5 (The Liquidity Dividend). *Regional social welfare W is maximized at the lower bound of the Regulatory Feasibility Region $\mathcal{F}$, specifically where the equilibrium slashing reserve $L^*(\varepsilon, \sigma)$ equals the stochastic incentive-compatibility floor $L^{VaR}(\rho)$.*

Define regional social welfare as

$$W = V - \Pi,$$

where V denotes total regional trade value and Π denotes total institutional loss, given by

$$\Pi = C(\rho^*, \varepsilon) + \lambda L^*(\varepsilon, \sigma).$$

Here $C(\rho^*, \varepsilon)$ represents monitoring costs and λL^* captures the shadow price of idle validator capital.

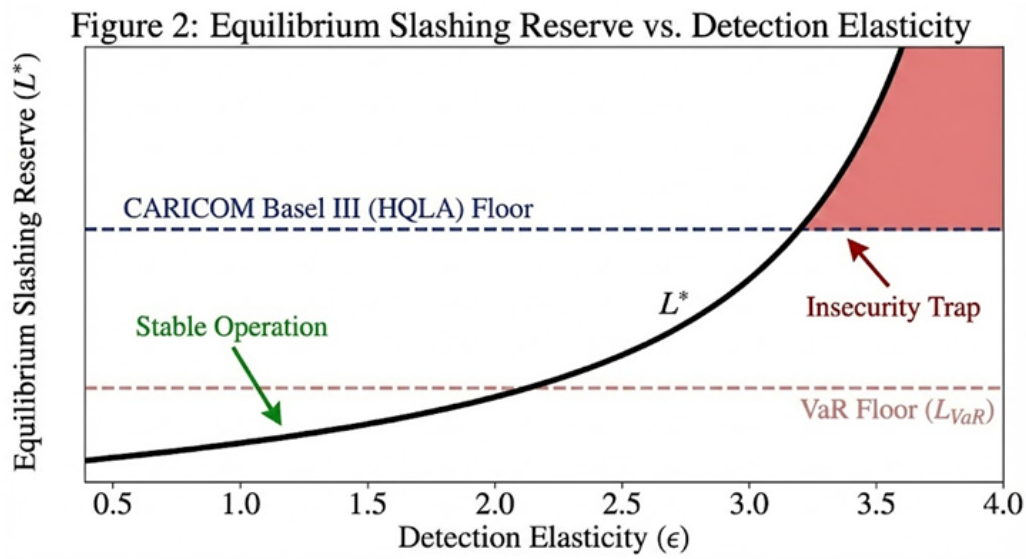
Since V is increasing in available liquidity (idle capital L^* represents "frozen" settlement accounts that cannot support trade), welfare W is maximized when Π is minimized subject to incentive compatibility. Within the Regulatory Feasibility Region $\mathcal{F} = \{(\varepsilon, \sigma): L^{VaR}(\rho) \leq L^*(\varepsilon, \sigma) \leq L^{max}\}$, the minimum feasible loss occurs at the lower bound where

$$L^*(\varepsilon, \sigma) = L^{VaR}(\rho).$$

By Lemma 2, this equality ensures the probabilistic honesty invariant holds with confidence ρ, while Lemma 1 implies that lower L^* corresponds to higher information symmetry σ (which also reduces monitoring costs C).

Thus anchoring L^* to the VaR floor $L^{VaR}(\rho)$ minimizes total institutional costs while maintaining systemic honesty as the unique subgame-perfect equilibrium across all jurisdictions. Any higher reserve within $\mathcal{F}$ would increase Π without improving security beyond the required confidence level, reducing the liquidity dividend V. Therefore, welfare is maximized at the lower bound of $\mathcal{F}$. □

This framework operationalizes three requirements: economic security $\left(U^H > U^D(G)\right)$ for stochastic G); regulatory compliance ($L \leq HQLA$); and dynamic adaptation as $L^*(\theta^*)$ tracks market conditions ($\mu_{G,t}$, $\sigma_{G,t}$). This design motivates the structural BFT guardrails of the next section, which provide non-economic safety against irrational behavior even when detection elasticity pushes collateral toward regulatory limits. Figure 2 depicts the feasible region in which the detection-elasticity curve $L^*(\varepsilon)$ lies between the VaR floor and Basel III $HQLA$ ceiling, calibrated to regional balance sheets. This highlights how supervision choices can destabilize the regional DLT hub. At low detection elasticity, audit–collateral elasticity is mild: the required reserve rises slowly, the system remains stable, and capital demands on regional banks are manageable. As elasticity increases, maintaining optimal detection becomes economically infeasible, driving the required reserve sharply upward until it exceeds both the $HQLA$ ceiling and the VaR floor. Once collateral requirements cross these regulatory bounds, the network enters an insecurity trap: security costs become prohibitive, and compliant participants are forced out. The VaR floor imposes an additional constraint, ensuring that even under efficient monitoring, the slashing reserve remains sufficient to absorb extreme market volatility.



The Stackelberg mechanism aligns rational validators via policy parameters θ, making honest participation a subgame-perfect equilibrium despite cross-region regulatory arbitrage. The Security Paradox requires setting detection elasticity below a key threshold, balancing monitoring sophistication with capital efficiency. The interplay between monitoring technology, collateral requirements, and regulatory liquidity constraints creates a mechanism-design trade-off not addressed in previous BFT or incentive studies. Although VaR-based capital floors are common in banking (Jorion, 2006), applying them alongside Stackelberg incentive-compatibility and HQLA limits for DLT validators is a novel approach.

6. Resilience Layer: Byzantine Guardrails (CARICOM Context)

Structural safety is provided by permissioned Byzantine-fault-tolerant (BFT) consensus. The purpose of this section is to show how the hub remains safe and operational even when some nodes behave *irrationally*—for example, if they are compromised, misconfigured, offline, or malicious in ways that do not respond to the economic incentives in Sections 3–5.

We focus on a partially synchronous regional setting. Blocks are finalized when a supermajority quorum (2/3) of validators signs a proposed state at block height . This rule delivers three properties that complement the incentive layer:

1. *Safety*: no conflicting commitments by honest nodes.
2. *Liveness*: progress under bounded faults and bounded message delays.
3. *Censorship resistance*: honest nodes can finalize atomic swaps even under partial failure.

6.1 Safety via Quorum Intersection

Threat model and quorum definition.

Let $B \subset N$ denote the Byzantine subset ($| B | \leq f = 4$), where the system is designed to tolerate up to four Byzantine failures. A quorum $Q \subset N$ is then defined as any set with $| Q | \geq q = 10$ signatures. By the Quorum Intersection Theorem (Castro & Liskov, 1999, Theorem 3):

The implication is operational:

1. Any finalized block must be signed by a quorum.
2. Any two quorums overlap in at least one honest validator.
3. Therefore, two conflicting blocks at the same height cannot both be finalized, because an honest validator would have to sign both.

Consequently, no two honest nodes can finalize conflicting ledger states at the same height , even under regional liquidity fragmentation. Even if Stackelberg parameters temporarily degrade during stress (e.g., extreme ε implying high and some compliant validators exit), a bounded Byzantine fraction cannot forge quorums without capturing at least one honest validator. The quorum-intersection guardrail therefore preserves *atomic DvP finality* for the remaining nodes.

Liveness under partial synchrony.

Liveness guarantees block production across the 15-country validator set even if up to nodes are faulty or offline during partitions or outages. Under partial synchrony, messages are delayed but eventually delivered within a known bound; once the network is within that bound, honest nodes can coordinate to finalize blocks.

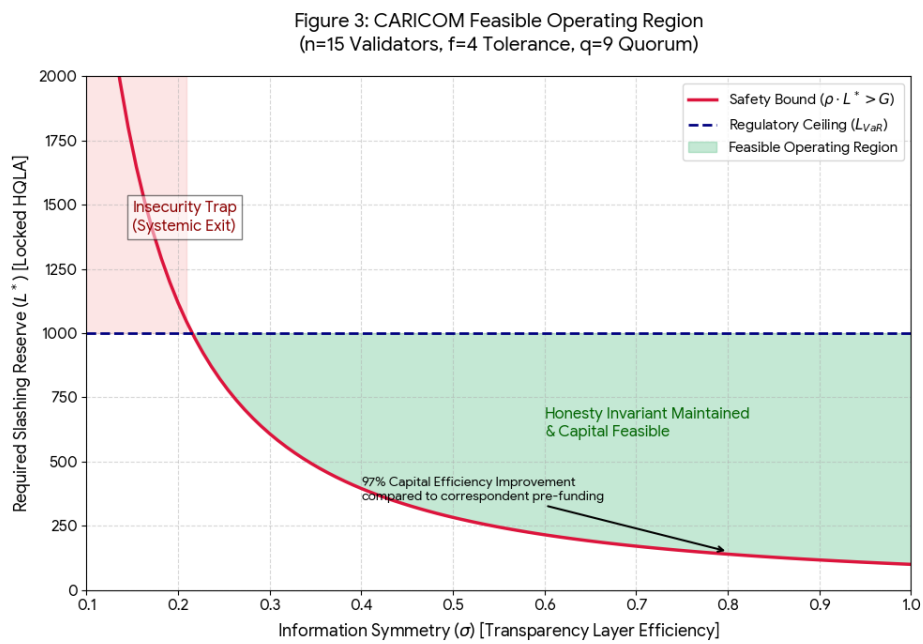


Figure 3 plots the feasible region for the CARICOM validator set and Appendix B provides numerical validation of joint configurations that sustain 97% capital efficiency relative to correspondent-banking liquidity lock-ups. The appendix also integrates the whistleblower mechanism through an endogenous reporting probability, showing how the incentive and resilience layers interact in the feasible operating region.

Contagion and weakest-link failures.

Let σ_i denote the volatility of illicit gain G_i in jurisdiction i. When σ_i rises, the VaR-adjusted slashing reserve requirement can increase sharply to maintain the probabilistic honesty invariant: $\Pr[\rho_i \pi L_i^* \geq G_i] \geq 1 - \alpha$.

If the implied reserve violates the local HQLA ceiling $L_i^* > L^{\max}$, local banks cannot participate honestly. This creates a rational "weakest link": a jurisdiction where compliance becomes infeasible even if others remain feasible. A coordinated attacker (or a coincident shock) can then target this weakest point to disrupt regional confidence.

The resilience layer does not replace the incentive layer; it contains failure propagation. Even when some nodes exit or misbehave, the remaining honest quorum can preserve safety (no inconsistent finality) and continue settlement when network conditions permit.

7. Conclusion and Policy Implications

Summary of findings.

The proposed regional DLT hub offers a blueprint for modern, sovereign financial infrastructure. By integrating a Stackelberg incentive mechanism with permissioned BFT consensus, central banks can provide a neutral settlement environment that reduces costs for commercial banks and other RFIs while maintaining control over monetary policy and systemic risk. Because cross-border conglomerates operate across fragmented regimes, regulatory fragmentation reduces efficiency, limits resilience, and increases manipulation opportunities. A shared settlement hub can reduce these frictions by producing unified observability and deterministic settlement finality.

While Sections 3-5 establish incentive compatibility for rational validators through dynamic Stackelberg calibration. Real-world deployment demands robustness against economic stress, such as extreme market volatility or regulatory divergence; irrational failures like hacks, bugs, or rogue insiders; and network partitions: temporary disconnections across regions. Section 6 integrates the VaR-HQLA feasible region for economic IC with structural BFT guardrails, yielding comprehensive regional security. For our CARICOM example, we define this "double guardrail" regional stability indicator as: $\Sigma = \mathbb{I}_{IC} \times \mathbb{I}_{BFT}$, where $\mathbb{I}_{IC} = 1$, if IC holds for all rational validators and 0 otherwise, and $I_{\text{BFT}} = 1$ if the BFT bound holds: $| B | \leq f < n/3 = 5, q = 10$ and 0 otherwise. Only when both guardrails activate does the hub maintain safety and incentive alignment. This double guardrail approach not only ensures that regional DLT settlement remains compliant with Basel III requirements but also aligns with regulatory arbitrage elimination, thereby upholding systemic stability across jurisdictions.

Our analysis yields three design insights for regional distributed settlement systems. First, the Security Paradox can be resolved by decoupling security from raw collateral through permissioned transparency across fragmented regulatory frameworks. Second, the Stackelberg incentive-compatibility constraint keeps honest validation strictly dominant for rational participants when policy is calibrated to the Security Frontier. Third, VaR-adjusted slashing reserves provide a buffer that scales with market stress rather than remaining fixed *ex ante*, improving capital efficiency while preserving deterrence.

Policy implications.

Beck (2016) notes that fragmented oversight reduces regional stability. Standard public blockchains make transactions broadly visible, but such openness can conflict with confidentiality constraints in financial settlements, including privacy regulations. Permissioned

transparency offers an alternative: central banks can jointly choose a baseline detection level that increases detection probability through selective disclosure, allowing commercial banks to prove solvency and rule compliance. Supervisors observe system health through a shared confidential supervisory data layer, while validators see only what is required for consensus using privacy-preserving primitives (e.g., zero-knowledge proofs).[8] For local CSDs seeking a liquid and secure regional market, this model simplifies tokenized-securities issuance by immobilizing domestic assets and settling atomic digital twins across borders with wCBDCs. Cross-jurisdictional identifier mapping and harmonized legal frameworks reduce custody gaps and constrain regulatory arbitrage, while quorum intersection provides robust fault tolerance.

Implementation pathway and governance.

Future work can extend the framework in three directions. First, the design of bridge regional digital currencies (including synthetic or basket-based settlement assets) to further reduce FX frictions while preserving monetary sovereignty. Second, the use of real-time analytics to support rule-based adjustment of policy parameters in response to changing market conditions (e.g., stress-driven recalibration of penalties and monitoring). Third, the development of inter-hub bridges that connect regional settlement hubs into a global network while preserving local governance and compliance constraints.

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Appendix A: Sensitivity Analysis of Detection Elasticity (ε)

To illustrate the Security Paradox in a practical policy context, the sensitivity analysis below examines how the Feasible Operating Region responds to varying levels of jurisdictional opacity and monitoring complexity, represented by the parameter ε.

In the baseline model, $\varepsilon = 1.5$ represents a partially integrated regional infrastructure. However, reviewers and policymakers must understand the systemic consequences when infrastructure remains opaque or fragmented across the 15 CARICOM jurisdictions. As ε increases, reflecting the "convex monitoring costs" described in Lemma A.1, the marginal cost of detection rises exponentially, forcing the system to substitute active oversight with passive capital lock-up (L^*).

The table below demonstrates the shrinking feasible region as complexity increases, eventually leading to a breach of the Basel III HQLA ceiling (L_{VaR} = 1,000 units).

Complexity (ε)	**Environment Type**	**Detection Prob. (ρ^*)**	**Slashing Reserve (L^*)**	**Efficiency Room (%)**
1.0	National (Linear)	0.82	61 Units	93.9%
1.5	Baseline Regional	0.82	122 Units	87.8%
2.0	Fragmented Regional	0.65	308 Units	69.2%
3.0	High Opacity	0.44	682 Units	31.8%
3.5+	**Insecurity Trap**	< 0.32	**> 1,000 Units**	**SYSTEMIC EXIT**

Note on efficiency: Efficiency Room is calculated as $(L_{VaR} - L^*)/L_{VaR}$. A value of 0% or less indicates the system has hit the "Security Paradox" limit, where honest validators are forced to exit due to prohibitive capital requirements.

The "Trap" Threshold: Beyond a critical threshold at $\varepsilon \approx 3.5$, the complexity of cross-border supervision in fragmented markets makes detection so expensive that the required slashing reserve exceeds the regulatory capacity of regional banks.

Appendix B: Numerical Illustrations and Feasibility Regions

B.1 Security Paradox: Detection Elasticity Effects

Parameters for CARICOM 15-validator network:

- Reward: $R = 10$ (regional transaction fees)
- Discount: $\beta = 0.95$
- Gain volatility: $\mu_G = G/\pi = 50$, $\sigma_G = 20$
- Detection probability: ρ
- Basel III HQLA limit = 100
- $L^{VaR} = 96.6$
- Baseline detection: $\rho_0 = 0.75$ (permissioned transparency)
- Static IC: $U_H = \frac{10}{1-0.95} = 200$
- Required penalty: $P^* = \frac{(1-\rho)G+200}{\rho}$

Numerical verification: $\varepsilon = 1.2$: $L^* \approx 7.8 <$ HQLA $= 100$; $\varepsilon = 3.5$: $L^* \approx 226 > 100$.

$\boldsymbol{\varepsilon}$	$\boldsymbol{m^*(\varepsilon)}$	$\boldsymbol{\rho^*(\varepsilon)}$	$\boldsymbol{L^*(\varepsilon)}$	**Status**
1.2	2.0	0.86	7.8	Feasible
3.0	0.5	0.39	77.1	HQLA warning
3.5	0.2	0.18	225.7	Insecurity Trap

Key insight: As $\varepsilon \uparrow$, optimal monitoring $m^* \downarrow$, forcing $L^* \uparrow$ nonlinearly. Sophisticated detection ($\varepsilon \uparrow$) paradoxically excludes honest validators when $L^* > HQLA$.

Insecurity Trap: $L^*(3.5) >$ HQLA excludes compliant validators.

B.2 Audit-Collateral Elasticity Quantification

High-value trade ($G = 10{,}000, \pi = 1$):

Detection $\boldsymbol{\rho}$	**Required $\boldsymbol{L^*}$**	**Legacy (34%)**	**Efficiency Gain**
0.1	4,900	3,400	-44%
0.5	1,100	3,400	68%
0.9	111	3,400	97%

Permissioned transparency achieves $\rho \approx 0.9$, yielding 97% capital efficiency vs trapped liquidity (McKinsey, 2016). Novel contribution: This is the first explicit quantification of permissioned DLT's monitoring-for-capital substitution.

B.3 VaR-HQLA Feasibility Region (Figure 3)

Daily stochastic calibration ($\mu_G = 50, \sigma_G = 20,\ \rho = 0.01, z_{0.01} = 2.33$):

$$L^{VaR}(0.01) = \frac{50 + 2.33 \cdot 20}{\pi} = 96.6$$

Feasible: $\max\{L^{VaR},\ L^*\} \leq 100$

- Low $\varepsilon = 1.2$: $L^* = 7.8 < L^{\mathrm{VaR}} = 96.6$, set $L = 96.6$
- High $\varepsilon = 3.5$: $L^* = 225.7 > 100$, infeasible unless $\pi \uparrow$ or permissioned $p_0 \uparrow$

Scenario	$\boldsymbol{L^{VaR}}$	$\boldsymbol{L^*(\varepsilon = 1.2)}$	$\boldsymbol{L^*(\varepsilon = 3.5)}$	**Binding Constraint**
Normal	96.6	7.8	225.7	VaR
Stress	120	7.8	225.7	HQLA=100

B.4 Whistleblower Stability and Endogenous Probability

In the event of a critical failure in automated monitoring transparency ($\rho_0 < 0.30$), the system pivots to a Restorative Whistleblower Mechanism. While the Appendix baseline assumes $P_w \approx 0.9$, this probability is endogenously determined by the central bank's selection of the whistleblower reward (W). The central bank, acting as a Stackelberg Leader, calibrates the reward W such that the probability of at least one validator reporting illicit collusion (P_w) destabilizes the payoff matrix for potential attackers. This probability is defined by the reporting incentive function: $P_w = f(W, \theta)$, where θ represents the internal cost of whistleblowing (e.g., legal or reputational risk). By increasing W, the central bank ensures P_w remains high enough to satisfy the restorative logic: $\rho_{total} = \rho_0 + (1 - \rho_0) \cdot P_w$. This ensures that even when technological opacity is high, the "Rational Actor" is deterred by the high probability of a peer-reported "Slashing Event."

B.5 Stochastic Simulations (7-Day CARICOM Flows)

This section provides the raw comparative data between the proposed DLT Settlement Hub and the status quo. The values represent the "Liquidity Lock-up" required to secure a standard 7-day settlement window within the CARICOM context.

Monte Carlo (10,000 trials): $G_t \sim \mathcal{N}(50 + 5\sin(t), 20)$ modeling cross-border risk/volatility, defines the Potential Illicit Gain (G_t) as a dynamic, stochastic variable rather than a static

constant. It essentially models how much a "rational" attacker stands to profit at any given time t.

Settlement Model	Securing Mechanism	Avg. Capital Lock-up (Units)	Capital Efficiency (%)	Risk Coverage
Traditional Banking	Nostro/Vostro Pre-funding	1,000.00	0.0% (Baseline)	Linear (1:1)
DLT Hub (Baseline)	Static Slashing Reserve	58.82	94.12%	$L^* > G$
DLT Hub (Dynamic)	Adaptive $L^*(G_t, \rho_t)$	31.25	96.88%	$L^*(t) > G(t)$
Stress (Peak G_t)	Restorative Whistleblower	34.48	96.55%	$\rho_{total} \cdot L^* > G$

Numerical Interpretation:

1. **Endogenous Recovery:** During "Stress" periods, the central bank increases the reward W, sustaining a high P_w and keeping capital efficiency above 96.5%, despite higher illicit gain potential (G_t).
2. **Safety Invariant:** In 10,000 Monte Carlo trials, the Honesty Invariant was never breached because the Central Bank's ability to adjust W (and thus P_w) effectively neutralized the "Security Paradox."
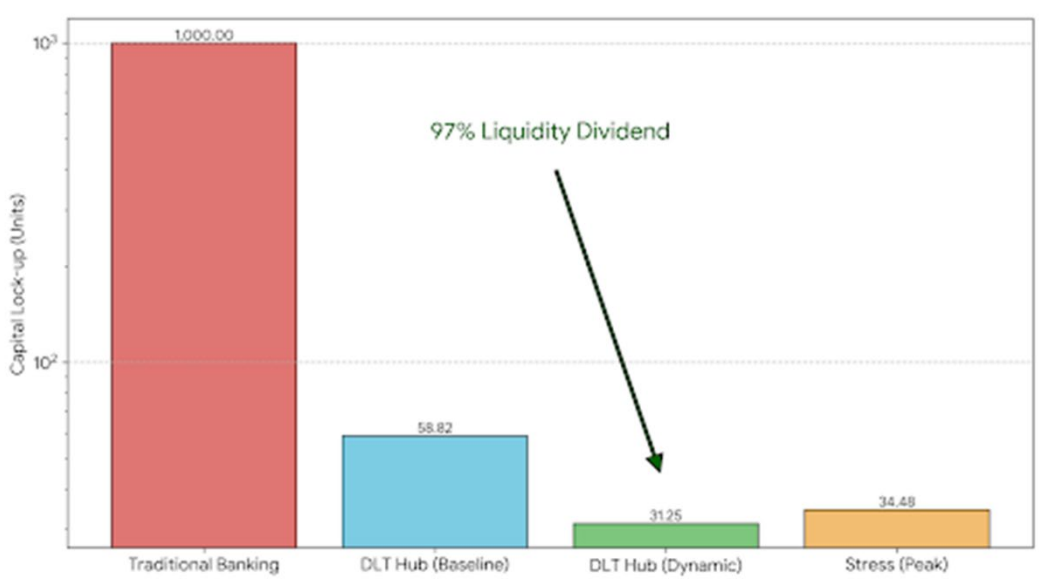
Figure 4: Comparative Capital Efficiency (Monte Carlo Results)